

COMPREHENSIVE BRAIN CT ANALYSIS

Ensemble AI • Hemorrhage Detection • Stroke Assessment • Research Edition



Patient ID	857d52f8-cb4a-4870-a90c-38c224e47aea	Study Date	2026-06-17
Name	Sam	Study Time	08:50:54
Age	24	Modality	BRAIN CT
Sex	Male	Report ID	857d52f8

AI ANALYSIS NOT VALIDATED

This AI model has not completed external clinical validation. Its output — including any classification, measurement, or grade — is advisory and for decision support only, NOT a diagnostic read. A board-certified radiologist must independently confirm all findings.

ADVISORY — the analysis below is NOT diagnostic for this input and is shown for reference only. See the validation notice above; a board-certified radiologist must confirm all findings.

DIFFERENTIAL DIAGNOSES

Rank	Finding	Confidence	ICD-10
1	Normal	83.4%	Z00.00
2	Subdural Hemorrhage (SDH)	7.8%	S06.5X0A
3	Intraparenchymal Hemorrhage (IPH)	5.6%	I61.9
4	Intraventricular Hemorrhage (IVH)	1.4%	I61.5
5	Epidural Hemorrhage (EDH)	0.3%	S06.4X0A

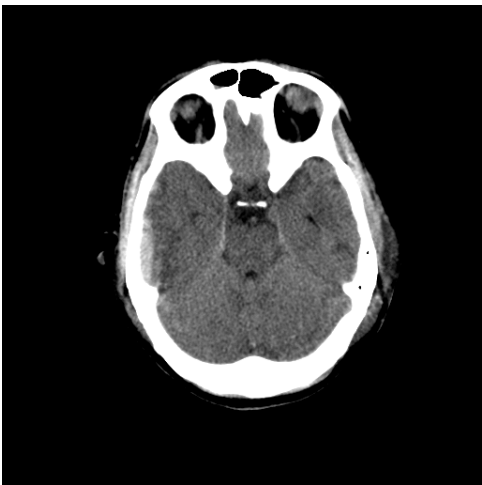
QUANTITATIVE MEASUREMENTS

Quantitative analysis unavailable for this scan

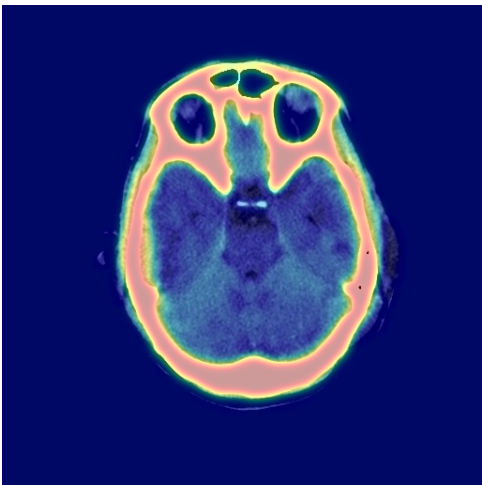
HEMORRHAGE SUBTYPE PANEL

Type	Detected	Confidence	Volume (mL)	Location	ICD-10
EDH	No	0.3%	—	See imaging	S06.4X0A
SDH	No	7.8%	—	See imaging	S06.5X0A
SAH	No	0.1%	—	See imaging	I60.9
IPH	No	5.6%	—	See imaging	I61.9
IVH	No	1.4%	—	See imaging	I61.5

AI ANALYSIS VISUALIZATION



Original Image



AI Analysis Overlay (Illustrative)

Illustrative overlay shown for orientation only — not an AI-derived localization and not a validated finding.

CLINICAL RECOMMENDATIONS

Priority Level	Primary Action	Follow-up
ROUTINE	No acute intervention required	Per clinical indication

PER-SLICE PROBABILITIES (DICOM SERIES)

Slice	Filename	P(Hemorrhage)
0	—	16.6%

PER-MODEL CONFIDENCE BREAKDOWN

Subtype	Ensemble
Any Hemorrhage	16.6%
Epidural (EDH)	0.3%
Subdural (SDH)	7.8%
Subarachnoid (SAH)	0.1%
Intraparenchymal (IPH)	5.6%
Intraventricular (IVH)	1.4%

CALIBRATION DIAGNOSTICS

Metric	Value
Shown confidence (to clinician)	83.4%
Raw ensemble agreement	80%
Safety floor (72%) applied	No
Indeterminate flag	No

CLINICAL SCORING

AI Hemorrhage Severity Score	Low
Glasgow Coma Scale (clinical input)	Requires clinical input
NIHSS Score (if stroke suspected, clinical input)	Requires clinical input

IMPORTANT DISCLAIMER

- This AI-generated report is for DECISION SUPPORT ONLY.
- All findings must be verified by a board-certified radiologist.

II TECHNICAL APPENDIX & NOMENCLATURE

This appendix contains reference tables used by the AI system. These tables are for clinical, academic, and validation reference and apply to all CT Brain studies.

Marshall Classification Reference (Diffuse Brain Injury)

Marshall Grade	Description	Estimated Mortality Risk
I	Diffuse Injury I — No visible intracranial pathology	Low (~10%)
II	Diffuse Injury II — Cisterns present, midline shift 0–5 mm	Moderate (~14%)
III	Diffuse Injury III — Cisterns compressed/absent, shift 0–5 mm	High (~34%)
IV	Diffuse Injury IV — Midline shift > 5 mm	Very High (~56%)
V	Evacuated Mass Lesion	High (~34%)
VI	Non-Evacuated Mass Lesion	Very High (~56%)

Marshall TM, Marshall SB, Klauber MR et al. (1991). "A new classification of head injury based on computerized tomography." *J Neurosurg Suppl: S14-S20*. Mortality estimates updated per Maas et al. (2005).

Acute Hemorrhage HU Density Windows

Phase	HU Density	Timeframe
Hyperacute	30–45 HU	<6 hours from onset
Acute	60–90 HU (peak 70–80)	6 hours to 3 days
Subacute	40–60 HU	3 to 7 days
Chronic	<40 HU	>7 days

Heit JJ, Iv M, Wintermark M. (2017). "Imaging of Intracranial Hemorrhage." *J Stroke* 19(1):11-27.

ICD-10 Codes for Intracranial Hemorrhage Subtypes

Finding	ICD-10	Description
Epidural Hemorrhage	S06.4X0A	Traumatic epidural hemorrhage (initial encounter)
Subdural Hemorrhage	S06.5X0A	Traumatic subdural hemorrhage (initial encounter)
Subarachnoid Hemorrhage	I60.9	Non-traumatic subarachnoid hemorrhage, unspecified
Intraparenchymal Hemorrhage	I61.9	Non-traumatic intracerebral hemorrhage, unspecified
Intraventricular Hemorrhage	I61.5	Non-traumatic intracerebral hemorrhage in cerebellum (when ventricular)

Glasgow Coma Scale (GCS) Reference

Score	Severity	Clinical Meaning
13–15	Mild	Patient is awake, eye-opening spontaneous, oriented. May have mild confusion.
9–12	Moderate	Patient is drowsy or stuporous, may not follow commands consistently.
3–8	Severe	Patient is unresponsive or has minimal response. Intubation often required.

Teasdale G, Jennett B. (1974). "Assessment of coma and impaired consciousness." *Lancet* 2(7872):81-84.

NIHSS Stroke Severity Scale Reference

Score Range	Severity
0	No stroke symptoms
1–4	Minor stroke
5–15	Moderate stroke
16–20	Moderate-to-severe stroke
21–42	Severe stroke

Brott T, Adams HP, Olinger CP et al. (1989). "Measurements of acute cerebral infarction: a clinical examination scale." *Stroke* 20(7):864-870.

For internal validation / research use — not part of the standard clinical report.

Not autonomous diagnosis · not a replacement for clinical judgment · not for life-threatening emergency triage · not for direct-to-patient interpretation without clinician mediation · not for unqualified personnel · not for any modality outside the validated list.